



BUYERS' GUIDE

ESTYLE

SMARTWATCHES: Complexity Simplified



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A smartwatch has become a frenzy that is sweeping the consumer electronics world, as people are clamouring to get their hands on the best fully-capable smartwatch. At an affordable price point, small form factor and impressive computing power, this hot toy is more than your regular activity tracker or GPS.

Coin-sized chipboard

Inside the elegantly-designed casing of the smartwatch lie about 1000 electronic components on a coin-sized PCB board. Components includes everything that a typical smartphone has, starting from microphone, power management unit, capacitive touch controller, graphic processor, central processing unit, signal processor and gyroscope. Since a smartwatch is in constant contact with the human body, it also has health-sensing components and sensors such as pulse oximeter and MEMS.

All vital information is displayed on the touchscreen, which is the size of a regular dial of the watch. Wireless connectivity features include Wi-Fi, NFC and GPS. And, the entire smartwatch works on the dedicated system-on-chip (SoC) principle.

The active ingredients

There are some active ingredients that go into the smartwatch that need to be understood.

Open source operating systems.

Android is the most popular platform in this segment. Built around Android operating system, watches based on it provide the standalone capability of Android smartphones.

In the long list of operating systems, there is Sailfish that bases itself on Linux. It is also found on bigger smart devices such as TVs and refrigerators.

Asteroid OS is an open source firmware replacement found in cost-effective smart wear devices focussed on community development.

Tizen was developed by Samsung for their smartwatch range, and Ubuntu Touch was developed by Canonical Ltd and Ubuntu community for smart devices.

WatchOS is a proprietary mobile operating system with open source components developed by Apple for their smartwatch platform, though it does not expose many features to the developer.

Development platform. The hardware runs on the development platforms created by developers. This forms the central unit on which the watch rests. Some key features that the developers are working on are wireless connectivity using Wi-Fi, Bluetooth and NFS using WICED. Balancing power consumption and performance while maintaining small form factor is done using platforms like Newton and Edison.

Sensors that stumble upon vital information. Sensors on a smartwatch can detect anything from heart rate and temperature to humidity, among others. Sensors that help accomplish this include motion sensors that detect acceleration and orientation using gyroscopes, geomagnetic sensors

Components of a smartwatch (Image courtesy: cellphones.com.vn)

